

**Risk-based Framework For Adaptive Quality Of Service And Threat-aware
Response In Software-defined Networking**

25-26J-028

Preliminary Progress Review

B.Sc. (Hons) Degree in Information Technology Specialized in Information
Technology

Department of Computer System and Network Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

January 2026

Student Details

Name: Perera K.C.S.P

Student ID: IT22324306

Topic: Threat detection Module

Supervisor: Mr. Uditha Dharmakeerthi

Co-Supervisor: Ms. Shashika Lokuliyana

Table of Contents

DECLARATION	Error! Bookmark not defined.
1. Introduction	1
1.1 Purpose	1
1.2 Definitions, Acronyms, and Abbreviations.....	1
1.3 Overview of the Document.....	1
2. Literature Review	2
2.1 Software-Defined Networking and Security Challenges.....	2
2.2 Traditional Intrusion Detection Systems in SDN.....	2
2.3 Machine Learning–Based Intrusion Detection in SDN	2
3. Literature Review	3
3.1 Background and Previous Work.....	3
3.2 Current State of the Research Problem	3
3.3 Identification and Significance of the Problem.....	3
4. Research Questions.....	4
5. Research Objectives.....	5
5.1 Main Objective	5
5.2 Specific Objectives	5
6. Research Methodology	6
6.1 Overall Methodological Approach	6
6.2 Conceptual Framework and Variables	7
6.3 Sources of Data and Data Collection.....	7
6.4 Data Analysis Methods	8
7. Preliminary Progress and Current Status.....	8
8. Sources for Test Data and Analysis	9
8.1 Sources for Test Data	9
8.2 Data Collection Procedures.....	9
8.2.1 Offline Data Collection	10
8.2.2 Online Data Collection	10
8.3 Data Analysis Methods	10
8.3.1 Machine Learning–Based Analysis.....	10
8.3.2 Performance Evaluation Metrics	10
9. Anticipated Benefits.....	11
9.1 Benefits to Users	11
9.2 Contribution to the Body of Knowledge	11

10. Scope and Expected Research Outcomes	12
11. Research Constraints	12
12. Project Plan and Schedule.....	12
13. References	13
13. Appendices.....	13

1. Introduction

1.1 Purpose

The purpose of this Preliminary Progress Review (PPR) Report is to formally document the current progress of the undergraduate research project titled *“Risk-Based Framework for Adaptive Quality of Service and Threat-Aware Response in Software-Defined Networking.”* This document is intended to provide supervisors, reviewers, and the academic evaluation panel with a clear understanding of the work completed to date, the methodology adopted, and the planned future activities.

This PPR does not describe the purpose of the research itself, but rather evaluates the **status, feasibility, and direction of the research project** at this intermediate stage of development.

1.2 Definitions, Acronyms, and Abbreviations

Acronym	Description
---------	-------------

SDN	Software-Defined Networking
-----	-----------------------------

QoS	Quality of Service
-----	--------------------

IDS	Intrusion Detection System
-----	----------------------------

ML	Machine Learning
----	------------------

DDoS	Distributed Denial of Service
------	-------------------------------

RF	Random Forest
----	---------------

PPR	Preliminary Progress Review
-----	-----------------------------

CICIDS2018 Canadian Institute for Cybersecurity Intrusion Detection Dataset 2018

1.3 Overview of the Document

This PPR report is organized into several sections. Following the introduction, a comprehensive literature review is presented, identifying existing work and research gaps. Research questions and objectives are then defined. The research methodology explains the technical and analytical approach adopted. The report also outlines the current progress, anticipated benefits, scope, constraints, project schedule, and references.

2. Literature Review

Software-Defined Networking (SDN) has emerged as a transformative paradigm in modern network architecture, fundamentally redefining how networks are designed, managed, and secured. By decoupling the control plane from the data plane, SDN introduces centralized programmability, global network visibility, and dynamic policy enforcement capabilities that are not feasible in traditional distributed networking models. While these characteristics significantly improve flexibility, scalability, and manageability, they also introduce **new security challenges** that require rethinking conventional network defense mechanisms.

2.1 Software-Defined Networking and Security Challenges

Early studies on SDN emphasize its architectural advantages, particularly centralized control, abstraction, and programmability. However, researchers have consistently highlighted that the **centralized SDN controller represents a single point of failure**. If compromised or overwhelmed, the controller can impact the entire network, making SDN environments especially vulnerable to attacks such as Distributed Denial of Service (DDoS), control-plane saturation attacks, and flow table exhaustion.

Unlike traditional networks where control intelligence is distributed across devices, SDN consolidates decision-making within the controller. This architectural shift increases the attack surface and amplifies the impact of malicious traffic if not detected and mitigated promptly. As a result, security has become a **primary concern in SDN research**, with intrusion detection and adaptive mitigation emerging as critical focus areas.

2.2 Traditional Intrusion Detection Systems in SDN

Intrusion Detection Systems (IDS) are a fundamental component of network security. Traditional IDS approaches are broadly classified into **signature-based** and **anomaly-based** systems. Signature-based IDS rely on predefined attack patterns and rules, making them effective against known threats but largely ineffective against zero-day attacks and evolving attack strategies. Anomaly-based IDS attempt to detect deviations from normal behavior but often suffer from high false positive rates.

When applied to SDN environments, traditional IDS solutions face additional limitations. Many operate as **external monitoring tools**, loosely coupled with the SDN controller. This separation limits real-time responsiveness and prevents seamless translation of detection results into immediate network control actions. Moreover, most traditional IDS solutions provide **binary outputs**, labeling traffic simply as benign or malicious, without considering the severity or potential impact of the detected threat.

2.3 Machine Learning–Based Intrusion Detection in SDN

To overcome the limitations of rule-based and signature-based IDS, recent research has increasingly explored the use of **Machine Learning (ML)** and **Deep Learning (DL)** techniques for intrusion detection in SDN. Supervised ML algorithms such as Random Forest, Support Vector Machines, and k-Nearest Neighbors have demonstrated strong performance on structured, flow-level network data. These models are particularly effective in learning complex, non-linear relationships between traffic features and attack behavior.

Deep learning approaches, including Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and hybrid CNN–LSTM models, have further improved detection accuracy by capturing temporal and spatial traffic patterns. Studies utilizing benchmark datasets such as **CICIDS2018** report detection accuracies exceeding 99% in controlled environments. However, these approaches often introduce high computational overhead, making them less suitable for real-time deployment within SDN controllers.

Despite their success, the majority of ML/DL-based IDS solutions share a common limitation: **their outputs are classification-oriented rather than risk-oriented**. Even when multi-class classification is used to identify specific attack types, the systems do not quantify the severity, urgency, or potential impact of detected threats.

3. Literature Review

3.1 Background and Previous Work

Software-Defined Networking (SDN) has emerged as a transformative networking paradigm by decoupling the control plane from the data plane and enabling centralized network control. This architectural shift allows dynamic policy enforcement, global traffic visibility, and enhanced programmability. However, the centralized SDN controller introduces new security vulnerabilities, making it an attractive target for cyberattacks.

Traditional Intrusion Detection Systems (IDS) rely heavily on signature-based or rule-based approaches, which are ineffective against zero-day attacks and evolving threat patterns. To overcome these limitations, recent research has focused on applying Machine Learning (ML) and Deep Learning (DL) techniques for intrusion detection in SDN environments.

Several studies have demonstrated that ML-based approaches such as Random Forest and Support Vector Machines provide fast detection, while DL models such as CNNs and LSTMs achieve higher accuracy by learning complex temporal patterns. Benchmark datasets such as CICIDS2018 are widely used for training and evaluation.

3.2 Current State of the Research Problem

Despite high detection accuracy, most existing SDN-based IDS solutions operate under a **binary classification paradigm**, categorizing traffic as either benign or malicious. Some systems adopt multi-class classification, identifying specific attack types. However, these approaches fail to quantify the **severity or risk level** of detected threats.

As a result, SDN controllers often apply rigid responses such as blocking traffic, which can degrade QoS and disrupt legitimate services. Minor attacks and large-scale attacks are treated equally, limiting the adaptive potential of SDN.

3.3 Identification and Significance of the Problem

The key research problem identified is the **absence of a severity-aware threat evaluation mechanism** within SDN security frameworks. The novelty of this research lies in introducing a **continuous threat scoring mechanism** that enables proportional and adaptive responses. This represents a significant technical contribution by bridging ML-based intrusion detection with real-time SDN policy enforcement.

4. Research Questions

This research is guided by a set of focused research questions that aim to investigate the feasibility, effectiveness, and practical implications of integrating severity-aware threat evaluation mechanisms into Software-Defined Networking (SDN) environments. These questions are formulated to address both the **technical challenges** and the **operational impact** of adopting a risk-based security approach within SDN.

RQ1: Severity-Aware Threat Scoring Using Machine Learning

How can machine learning techniques be effectively utilized to assign severity-aware threat scores to network flows in an SDN environment?

This research question explores the application of supervised machine learning models to analyze flow-level network features and generate probabilistic threat assessments. The focus is on determining how ML outputs can be transformed into continuous threat scores that reflect the **intensity, likelihood, and potential impact** of malicious activity, rather than relying solely on binary or categorical classifications.

RQ2: Impact of Continuous Threat Scoring on Adaptive Policy Enforcement and QoS

How does continuous threat scoring improve adaptive policy enforcement and Quality of Service (QoS) preservation when compared to traditional binary intrusion detection approaches?

This question investigates whether severity-based threat evaluation enables SDN controllers to apply **proportional and context-aware responses**, such as rate limiting, rerouting, or selective blocking. It aims to evaluate the extent to which continuous threat scoring reduces false positives, minimizes unnecessary service disruption, and maintains acceptable QoS levels during both normal operation and attack scenarios.

RQ3: Real-Time Integration and Practical Feasibility in SDN Controllers

How effectively can a threat evaluation component be integrated into an SDN controller to support real-time decision-making without imposing significant performance overhead?

This question addresses the practical feasibility of embedding a machine learning-based threat scoring module directly within the SDN control plane. It examines challenges related to real-time flow analysis, controller latency, scalability, and resource utilization, with the goal of ensuring that the proposed solution can operate efficiently in dynamic network environments.

5. Research Objectives

The objectives of this research are formulated to systematically address the identified research problem and to guide the development, implementation, and evaluation of a risk-based security framework for Software-Defined Networking (SDN). The objectives are structured into a **main objective** and a set of **specific objectives**, ensuring clarity, focus, and alignment with the overall research aim.

5.1 Main Objective

The primary objective of this research is to **design and implement a risk-based threat evaluation framework within a Software-Defined Networking environment** that is capable of assigning **dynamic, severity-aware threat scores** to network traffic flows and enabling **adaptive, Quality of Service (QoS)–aware security responses**. This objective aims to move beyond traditional binary intrusion detection models by providing actionable intelligence that supports proportional and context-aware network control decisions.

5.2 Specific Objectives

To achieve the main objective, the study seeks to accomplish the following specific objectives:

- **To critically analyze existing SDN-based intrusion detection systems** in order to identify their limitations, particularly with respect to binary classification, lack of severity assessment, and limited integration with adaptive network control mechanisms.
- **To design and develop a supervised machine learning–based intrusion detection model**, trained using the CICIDS2018 benchmark dataset, capable of accurately identifying malicious network flows based on flow-level traffic features.
- **To transform probabilistic outputs generated by the machine learning model into continuous threat scores**, enabling quantification of threat severity on a numerical scale and facilitating risk-aware decision-making within the SDN controller.
- **To integrate the trained machine learning model and threat scoring mechanism into an SDN control framework**, ensuring real-time flow inspection, efficient threat evaluation, and seamless communication with adaptive policy enforcement components.
- **To evaluate the performance of the proposed framework using both security-related and network-level metrics**, including accuracy, precision, recall, F1-score, false positive rate, latency, throughput, and scalability, in order to assess its effectiveness and practicality in realistic SDN environments.

This section presents a comprehensive description of the **research design, technical approach, and analytical procedures** adopted to achieve the objectives of this study. The

methodology has been carefully designed to be **systematic, reproducible, and scientifically rigorous**, while also ensuring practical applicability within real-world Software-Defined Networking (SDN) environments. The adopted approach integrates **machine learning–based threat detection** with **SDN-enabled adaptive control**, thereby enabling risk-aware and Quality of Service (QoS)–preserving security mechanisms.

The methodology follows best practices established in prior SDN security and intrusion detection research, combining offline analytical modeling with online system deployment and evaluation.

6. Research Methodology

6.1 Overall Methodological Approach

The research adopts an **experimental, implementation-driven methodological approach** that emphasizes both **analytical accuracy** and **operational feasibility**. The methodology is structured into two major phases:

Phase 1: Offline Machine Learning Model Development

In this phase, machine learning models are developed and trained using benchmark intrusion detection datasets. The offline nature of this phase allows computationally intensive tasks such as feature analysis, model training, and hyperparameter tuning to be conducted without imposing overhead on live network operations.

Key activities in this phase include:

- Dataset exploration and preprocessing
- Feature selection based on SDN-accessible flow statistics
- Training supervised machine learning models
- Evaluating detection performance and false positive rates
- Exporting the trained model for deployment

Phase 2: Online SDN Integration and Real-Time Evaluation

In the second phase, the trained model is integrated into an SDN controller to enable **real-time threat evaluation**. Incoming network flows are analyzed at the control plane, threat scores are computed dynamically, and adaptive policies are enforced based on severity levels.

This two-phase approach ensures a clear separation between **model learning** and **model inference**, which is critical for maintaining real-time performance and controller stability in SDN environments.

6.2 Conceptual Framework and Variables

The proposed conceptual framework consists of **four tightly integrated functional modules**, collectively enabling adaptive, severity-aware network security:

1. **QoS Scoring Module:** This module continuously evaluates the Quality of Service characteristics of network flows using performance-related metrics such as latency, packet loss, jitter, and throughput. The QoS score provides insight into the operational state of the network and supports balanced decision-making.
2. **Threat Scoring Module:** The threat scoring module applies the trained machine learning model to flow-level features extracted by the SDN controller. The model outputs probabilistic predictions, which are converted into **continuous threat scores** representing the severity and potential impact of detected anomalies.
3. **Adaptive Policy Engine:** This engine serves as the decision-making component of the framework. It combines threat scores and QoS scores to determine appropriate mitigation actions, such as allowing traffic, rate limiting, rerouting, or blocking malicious flows.
4. **Critical Mode Response Module:** The critical mode is activated when high-severity threat scores exceed predefined thresholds. In this mode, restrictive policies are enforced to protect essential services and prevent network-wide degradation.

Variables Identification

- **Independent Variables:** Flow-level network features including packet count, flow duration, byte rate, protocol type, source and destination ports, and flag statistics.
- **Dependent Variable:** The computed **threat score**, which quantifies the severity of network flows on a numerical scale and directly influences policy enforcement decisions.

6.3 Sources of Data and Data Collection

The primary data source for this research is the **CICIDS2018 dataset**, a widely recognized benchmark dataset developed by the Canadian Institute for Cybersecurity. The dataset includes realistic network traffic scenarios, encompassing both benign activity and a wide range of attack types such as DDoS, port scanning, brute-force attacks, and botnet communication.

Data collection is conducted through two complementary approaches:

Offline Dataset Collection

- Flow-level records from CICIDS2018 are used to train and validate the machine learning model.
- Data preprocessing steps include handling missing values, removing infinite values, feature normalization, and label encoding.

- The dataset is partitioned into training, validation, and testing subsets to ensure unbiased performance evaluation.

Online SDN Simulation Data

- Additional flow statistics are collected from SDN-based network simulations.
- Network emulation tools are used to generate controlled traffic scenarios, including both normal and attack conditions.
- These simulations provide real-time flow data for validating the deployed threat scoring framework.

6.4 Data Analysis Methods

The core data analysis technique employed in this study is **supervised machine learning–based classification**, with a **Random Forest classifier** selected as the primary detection model. Random Forest is chosen due to its suitability for structured tabular data, robustness against overfitting, and ability to produce probabilistic outputs essential for threat scoring.

The trained model is evaluated using standard security metrics, including:

- Accuracy
- Precision
- Recall
- F1-score
- False Positive Rate

Beyond traditional classification evaluation, the model’s probabilistic outputs are mapped to a **continuous threat score range of 0–10**, enabling severity-aware analysis. This threat score serves as the key input to the adaptive policy engine, allowing the SDN controller to apply proportional responses based on the assessed risk level.

Additionally, system-level performance metrics such as controller latency, throughput, and scalability are considered during the SDN integration phase to assess the practical feasibility of the proposed framework.

7. Preliminary Progress and Current Status

This section provides a detailed account of the work completed so far in relation to the proposed research plan. It highlights the milestones achieved, tools and technologies used, and the current maturity level of each project component.

At the time of this PPR submission, the following tasks have been completed:

- Selection and understanding of the CICIDS2018 dataset.
- Development of a complete ML training pipeline using a Jupyter Notebook.
- Training and evaluation of a Random Forest-based intrusion detection model.
- Exporting the trained model for deployment.
- Development of threat scoring logic for converting ML probabilities into severity scores.

The ML component of the project is approximately **85–90% complete**, representing a significant milestone.

8. Sources for Test Data and Analysis

This section describes the **sources of test data**, the **procedures used for data collection**, and the **methods applied for data analysis** in order to validate the proposed risk-based threat evaluation framework. The selection of data sources and analytical techniques is guided by the need for **realism, reproducibility, and relevance** to Software-Defined Networking (SDN) environments.

8.1 Sources for Test Data

The primary source of test data for this research is the **CICIDS2018 dataset**, developed by the Canadian Institute for Cybersecurity. This dataset is widely recognized in the research community for intrusion detection and network security studies due to its realistic traffic generation, comprehensive feature set, and inclusion of diverse modern attack scenarios.

The CICIDS2018 dataset includes:

- Benign network traffic representing normal user behavior
- Multiple attack types such as Distributed Denial of Service (DDoS), port scanning, brute-force attacks, botnet communication, and infiltration attempts
- Flow-level features extracted using CICFlowMeter, which align well with the statistics available from SDN controllers

In addition to the benchmark dataset, **synthetic network traffic** generated through SDN simulations is used as a secondary data source. This traffic is produced in controlled environments to evaluate the behavior of the proposed framework under real-time conditions.

8.2 Data Collection Procedures

Data collection in this research is conducted using a two-stage approach to support both offline model development and online system evaluation.

8.2.1 Offline Data Collection

- Flow-level CSV files from the CICIDS2018 dataset are collected and aggregated to form a comprehensive training and testing dataset.
- Data preprocessing procedures are applied, including:
 - Removal of missing and infinite values
 - Normalization of numerical features
 - Encoding of categorical variables
 - Handling of class imbalance through sampling techniques where necessary
- The processed dataset is partitioned into training, validation, and testing subsets to ensure unbiased performance evaluation.

8.2.2 Online Data Collection

- SDN-based network simulations are conducted using network emulation tools to generate live traffic flows.
- Flow statistics are collected directly from the SDN controller during network operation.
- Both benign and attack scenarios are introduced to evaluate the system's real-time detection and response capabilities.

This dual data collection strategy ensures that the model is trained on standardized benchmark data while being validated under realistic operational conditions.

8.3 Data Analysis Methods

The collected data is analyzed using **supervised machine learning techniques**, supported by statistical performance evaluation methods.

8.3.1 Machine Learning–Based Analysis

- A Random Forest classifier is used to analyze flow-level network features and detect malicious behavior.
- The model generates probabilistic outputs indicating the likelihood of a network flow being malicious.
- These probabilities are converted into **continuous threat scores** on a numerical scale ranging from 0 to 10, enabling severity-aware threat evaluation.

8.3.2 Performance Evaluation Metrics

The effectiveness of the proposed framework is evaluated using the following metrics:

- **Accuracy** to measure overall detection correctness

- **Precision** to assess the proportion of correctly identified malicious flows
- **Recall** to evaluate the ability to detect actual attacks
- **F1-score** to balance precision and recall
- **False Positive Rate** to measure the impact on legitimate traffic

In addition to security metrics, **network-level performance metrics** such as latency, throughput, and scalability are analyzed during SDN simulations to assess the operational impact of the proposed system.

9. Anticipated Benefits

9.1 Benefits to Users

The proposed framework offers several practical benefits to network administrators, system operators, and organizations managing SDN-based infrastructures.

Firstly, the introduction of **severity-aware threat scoring** enables administrators to move beyond traditional binary alerts and gain a more nuanced understanding of network threats. Instead of simply identifying whether traffic is malicious or benign, the system provides **quantitative threat scores** that indicate the severity and potential impact of each detected anomaly. This allows administrators to prioritize critical threats and respond more effectively.

Secondly, the framework contributes to a **reduction in false positives**, a common limitation of many intrusion detection systems. By incorporating probabilistic threat evaluation and contextual analysis, the system minimizes unnecessary blocking of legitimate traffic, thereby preserving service availability and reducing operational disruptions.

Furthermore, the integration of threat scoring with an **adaptive policy enforcement mechanism** enables real-time, proportional responses such as rate limiting, traffic rerouting, or selective blocking. This adaptive behavior ensures that security actions are applied in a manner that maintains **Quality of Service (QoS)** for legitimate users, even during attack scenarios.

Finally, the centralized visibility provided by SDN, combined with machine learning-based analytics, enhances **situational awareness** and supports informed decision-making. Network administrators benefit from improved monitoring, faster incident response, and greater confidence in the resilience of the network infrastructure.

9.2 Contribution to the Body of Knowledge

From an academic perspective, this research contributes to the evolving field of SDN security by addressing a critical gap in existing intrusion detection and response

mechanisms. While prior studies have focused primarily on improving detection accuracy using machine learning and deep learning techniques, this research shifts the focus toward **risk-aware and severity-based threat evaluation**.

The key contribution lies in the integration of **continuous threat scoring** with SDN control mechanisms, enabling adaptive and context-aware security enforcement. By demonstrating how probabilistic ML outputs can be transformed into actionable threat scores and embedded directly within the SDN control plane, the research advances existing approaches that rely on binary or multi-class classification.

Additionally, the proposed framework provides a **practical reference architecture** for future research in adaptive network security. The combination of offline machine learning training, real-time SDN integration, and severity-driven policy enforcement offers a reusable methodology that can be extended to other network environments and security applications.

Overall, this research contributes new insights into how **risk-based decision-making can enhance the effectiveness, flexibility, and resilience of SDN security systems**, thereby enriching the body of knowledge in network security and intelligent network management.

10. Scope and Expected Research Outcomes

The system will detect and score threats at the flow level and enforce adaptive SDN policies. Payload inspection and encrypted traffic decryption are outside the scope. Expected outcomes include a functional SDN-integrated threat scoring framework.

11. Research Constraints

Constraints include computational resource limitations, dataset imbalance, and real-time processing overhead within the SDN controller.

12. Project Plan and Schedule

A detailed project plan is essential to demonstrate the feasibility of the proposed research and to ensure that all project milestones can be achieved within the allocated time frame. The project is planned over a period of twelve months and is divided into clearly defined phases, each with specific deliverables.

The project follows a 12-month schedule:

- Months 1–3: Literature review and data preprocessing
- Months 4–6: ML model development (Completed)
- Months 7–9: SDN integration and adaptive policy engine
- Months 10–12: Testing, evaluation, and documentation

13. References

1. D. Kreutz, F. M. V. Ramos, P. E. Verissimo, C. E. Rothenberg, S. Azodolmolky and S. Uhlig, "Software-Defined Networking: A Comprehensive Survey," *Proceedings of the IEEE*, vol. 103, no. 1, pp. 14-76, Jan. 2015, doi: 10.1109/JPROC.2014.2371999.
2. I. Sharafaldin, A. H. Lashkari and A. A. Ghorbani, "Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization," *4th International Conference on Information Systems Security and Privacy (ICISSP)*, Funchal, Madeira, Portugal, 2018, pp. 108–116, doi: 10.5220/0006639801080116.
3. A. Moubayed, A. Refaey, H. Shami, and A. Shami, "Software Defined Perimeter (SDP): State of the Art Secure Solution for Modern Networks," *IEEE Network*, vol. 33, no. 5, pp. 226-233, Sept.–Oct. 2019, doi: 10.1109/MNET.001.1800523.
4. P. Idhammad, K. Afdel, and M. Belouch, "Detection System of HTTP DDoS Attacks in a Cloud Environment Based on Information Theoretic Entropy and Random Forest," *Security and Communication Networks*, vol. 2018, Article ID 1263123, pp. 1–13, 2018, doi: 10.1155/2018/1263123.

13. Appendices

The screenshot displays a Microsoft Excel spreadsheet titled "Friday-WorkingHours-Afternoon-Dropscap-ISC...". The spreadsheet contains a large table of data with columns labeled A1 through Z1. The data appears to be network-related, with columns for various metrics and rows of data. The spreadsheet is open in the Microsoft Excel application, showing the ribbon with tabs for File, Home, Insert, Draw, Page Layout, Formulas, Data, Review, View, Automate, Developer, and Help. The status bar at the bottom indicates "Ready Sheet 1 of 1" and "Accessibility: Unavailable".

Figure 0. Inside of dataset

Name	Date modified	Type	Size
Friday-WorkingHours-Afternoon-DDos.p...	1/1/2026 5:07 PM	Microsoft Excel Co...	75,317 KB
Friday-WorkingHours-Afternoon-PortSca...	1/1/2026 5:07 PM	Microsoft Excel Co...	75,104 KB
Friday-WorkingHours-Morning.pcap_ISC...	1/1/2026 5:07 PM	Microsoft Excel Co...	56,950 KB
Monday-WorkingHours.pcap_ISCX.csv	1/1/2026 5:07 PM	Microsoft Excel Co...	172,782 KB
Thursday-WorkingHours-Afternoon-Infilt...	1/1/2026 5:07 PM	Microsoft Excel Co...	81,155 KB
Thursday-WorkingHours-Morning-WebAt...	1/1/2026 5:07 PM	Microsoft Excel Co...	50,804 KB
Tuesday-WorkingHours.pcap_ISCX.csv	1/1/2026 5:07 PM	Microsoft Excel Co...	131,914 KB
Wednesday-workingHours.pcap_ISCX.csv	1/1/2026 5:07 PM	Microsoft Excel Co...	219,890 KB

Figure 0.3all datasets used

```

File Edit Selection View Go Run Terminal Help
C:\Users> sanj > OneDrive > Desktop > Research > PPI > QOS > QOScsp > threat detection model > project > models.ipynb > rf = RandomForestClassifier(
Generate + Code + Markdown Interrupt Restart Clear All Outputs Go To Jupyter Variables Outline Python 3.13.7

[12] ✓ 21.7s
...
((1289055, 69),
(376053, 69),
array([1026307, 182748]),
array([207348, 168705]))

rf = RandomForestClassifier(
    n_estimators=300,
    n_jobs=-1,
    random_state=RANDOM_SEED,
    class_weight="balanced"
)
rf.fit(Xtr, ytr)

# calibrate to get a reliable ThreatScore in [0,1]
cal = calibratedClassifierCV(rf, method="isotonic", cv=3).fit(Xtr, ytr)
proba = cal.predict_proba(Xte)[:,-1]
pred = (proba >= 0.5).astype("int8")

print(classification_report(yte, pred, digits=4))
print("ROC-AUC:", roc_auc_score(yte, proba))
print("PR-AUC :", average_precision_score(yte, proba))

# ROC
fpr, tpr, _ = roc_curve(yte, proba)
plt.figure(); plt.plot(fpr, tpr); plt.xlabel("FPR"); plt.ylabel("TPR"); plt.title("ROC"); plt.grid(True); plt.show()

# Precision Recall

```

Figure 0.2model is running